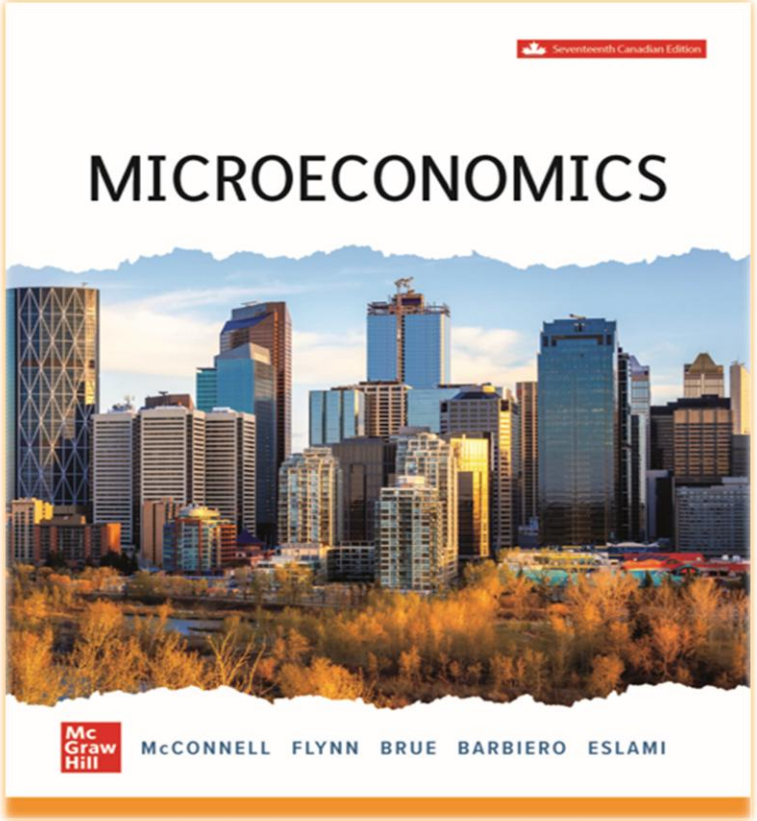
CHAPTER 4

Market Failures Caused by Externalities and Asymmetric Information



PowerPoint Presentation Prepared by:
Adian McFarlane, King's University College at Western University

Edit: Rashed Ahmed

© 2025 McGraw Hill Limited

LEARNING OBJECTIVES

LO4.1 Explain consumer surplus, producer surplus, and how properly functioning markets maximize total surplus and allocate resources optimally.

LO4.2 Explain how positive and negative externalities cause under- and overallocations of resources.

LO4.3 Explain why society is usually unwilling to pay the costs of completely eliminating negative externalities, such as air pollution.

LO4.4 Understand why asymmetric information may justify government intervention in some markets.

© 2025 McGraw Hill Limited

INTRODUCTION

- Properly functioning markets allocate resources efficiently.
- Externalities and asymmetric information are key causes of market failure.
- Market failures lead to underproduction, overproduction, or no production of goods and services.
- Government intervention, like pollution taxes, can help restore market efficiency when failures occur.

© 2025 McGraw Hill Limited

4.1 EFFICIENTLY FUNCTIONING MARKETS 1/7

- Competitive markets usually allocate resources efficiently but have exceptions where they fail.
- Market failure occurs when competition alone doesn't guarantee the highest net benefits from limited resources.
- Efficient markets require demand to reflect full consumer willingness to pay and supply to include all production costs.
- Total surplus, the sum of consumer and producer surplus, is maximized when markets function efficiently, benefiting society the most.

© 2025 McGraw Hill Limited

4.1 EFFICIENTLY FUNCTIONING MARKETS 2/7

Consumer Surplus

- Difference between what a consumer is willing to pay for a good and what the consumer actually pays.
- Extra benefit from paying less than the maximum price
- The maximum price that a person is willing to pay for a unit of a product depends on the opportunity cost of his consumption alternatives.
- The concept of maximum willingness to pay also gives us another way to understand demand curves.

© 2025 McGraw Hill Limited

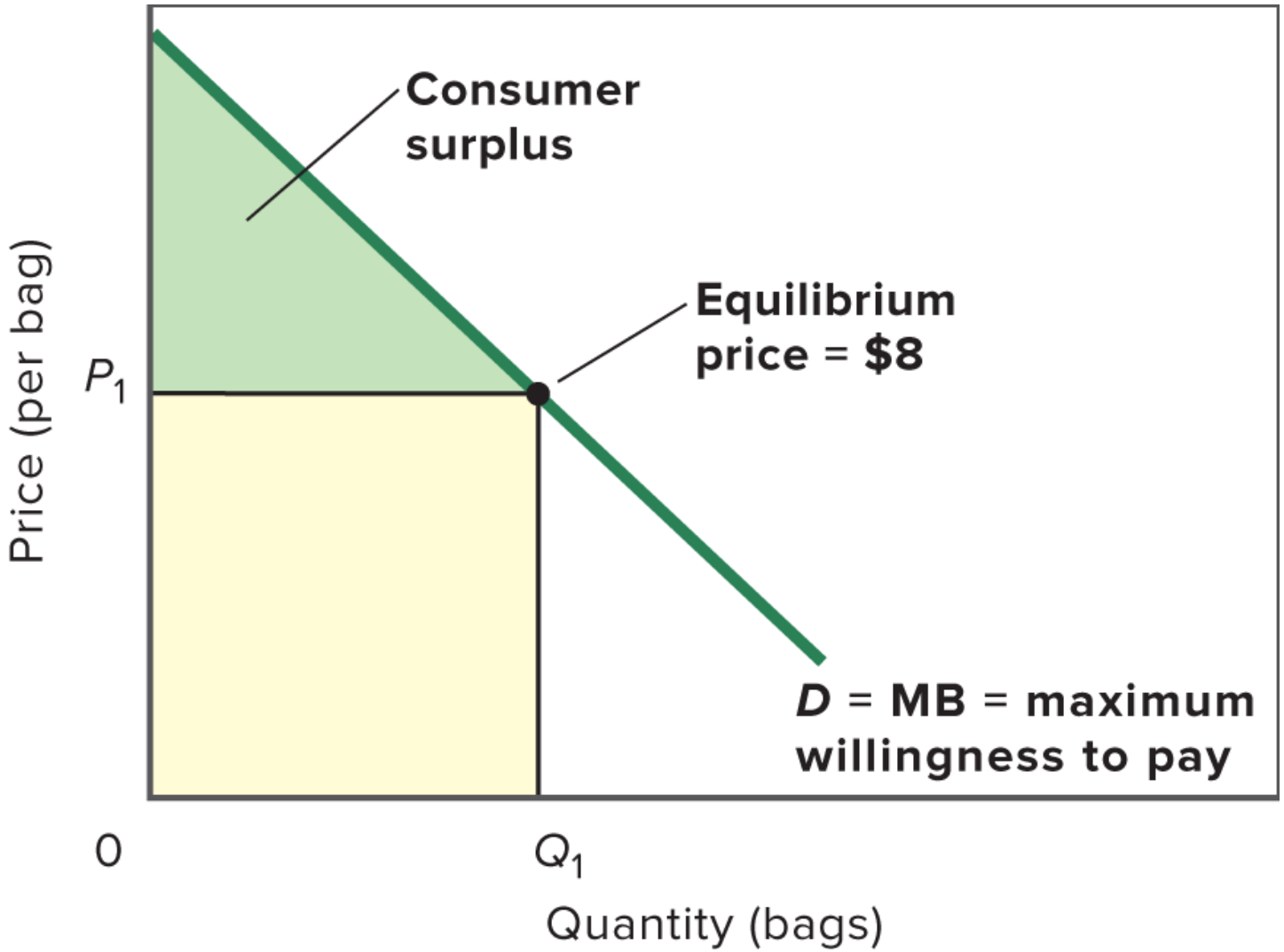
TABLE 4-1 Consumer Surplus

(1) Person	(2) Maximum Price Willing to Pay	(3) Actual Price (Equilibrium Price)	(4) Consumer Surplus
Bashir	$13	$8	$5 (=$13-$8)
Badal	12	8	4 (=$12-$8)
Bangshik	11	8	3 (=$11-$8)
Bart	10	8	2 (=$10-$8)
Bernis	9	8	1 (= $9-$8)
Betty	8	8	0 (= $8-$8)

© 2025 McGraw Hill Limited

FIGURE 4-1 Consumer Surplus

- Consumer surplus is the gap between willingness to pay and the equilibrium price.
- It's shown by the green triangle, representing the extra benefit at quantity Q_1.
- Consumers pay less than their maximum, shown by the green triangle.



© 2025 McGraw Hill Limited

4.1 EFFICIENTLY FUNCTIONING MARKETS 3/7

Producer Surplus

- Like consumers, producers also receive a surplus benefit from participating in markets.
- Difference between the actual price a producer receives and the minimum price they would accept → equal to a producer's marginal cost
- A producer's minimum acceptable price can also be interpreted as the opportunity cost of bidding resources away from producing other products.
- Extra benefit from receiving a higher price.

© 2025 McGraw Hill Limited

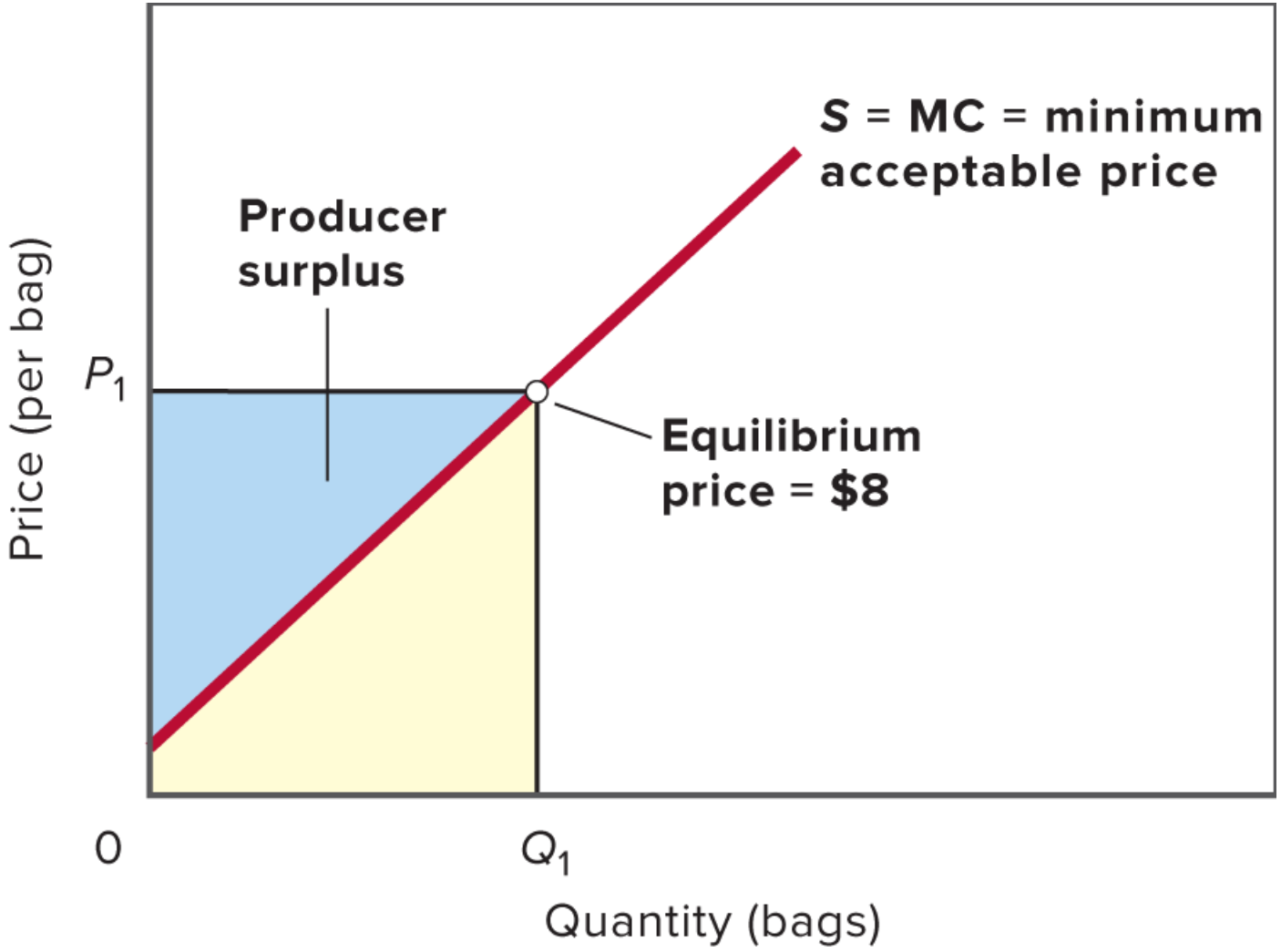
TABLE 4-2 Producer Surplus

(1) Person	(2) Minimum Acceptable Price	(3) Actual Price (Equilibrium Price)	(4) Producer Surplus
Carlos	$3	$8	$5 (=$8-$3)
Courtney	4	8	4 (=$8-$4)
Constantine	5	8	3 (=$8-$5)
Carolina	6	8	2 (=$8-$6)
Craig	7	8	1 (=$8-$7)
Chad	8	8	0 (=$8-$8)

© 2025 McGraw Hill Limited

FIGURE 4-2 Producer Surplus

- Producer surplus is the gap between the price received and the minimum acceptable price.
- The blue triangle represents this extra benefit at quantity Q_1.
- Surplus arises as producers get more than their minimum price.



© 2025 McGraw Hill Limited

4.1 EFFICIENTLY FUNCTIONING MARKETS 4/7

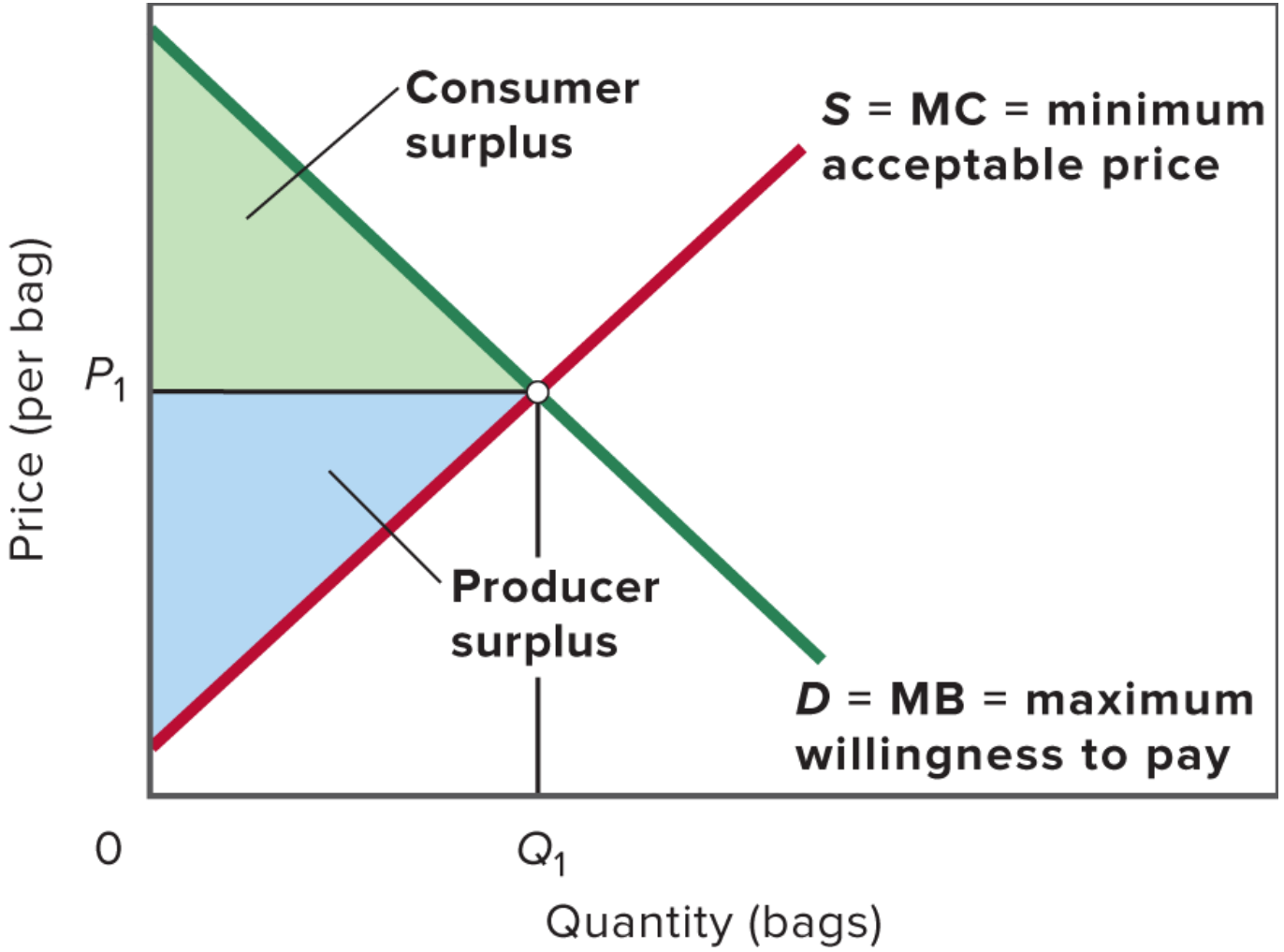
Total Surplus and Efficiency

- Figure 4-3 illustrates equilibrium price and quantity, highlighting consumer and producer surplus.
- Markets with down-sloping demand and up-sloping supply curves create consumer and producer surplus.
- Equilibrium reflects economic efficiency, combining productive efficiency (lowest-cost production) and allocative efficiency (optimal resource allocation).
- Productive efficiency minimizes costs, while allocative efficiency ensures resources are used where they have the highest value.

© 2025 McGraw Hill Limited

FIGURE 4-3 Efficiency: Maximum Combined Consumer and Producer Goods

- At Q_1, total surplus is maximized.
- Efficiency occurs when willingness to pay equals the minimum price.
- The blue and green triangles show maximized surpluses.



© 2025 McGraw Hill Limited

4.1 EFFICIENTLY FUNCTIONING MARKETS 5/7

Three Equivalent Conditions for Allocative Efficiency

1. Marginal Benefit = Marginal Cost

- Efficient resource allocation occurs when MB equals MC.
- Demand curves represent marginal benefit (MB), and supply curves represent marginal cost (MC).
- At equilibrium (Q_1), where demand and supply intersect, the output is allocatively efficient.

© 2025 McGraw Hill Limited

4.1 EFFICIENTLY FUNCTIONING MARKETS 6/7

Three Equivalent Conditions for Allocative Efficiency

2. Maximum Willingness to Pay = Minimum Acceptable Price

- Equilibrium (Q_1) is allocatively efficient when maximum willingness to pay equals minimum acceptable price.
- Up to Q_1, utility from production and consumption exceeds that of alternative uses.
- At Q_1, all opportunities where marginal benefits exceed marginal costs are fully utilized.

© 2025 McGraw Hill Limited

4.1 EFFICIENTLY FUNCTIONING MARKETS 7/7

Three Equivalent Conditions for Allocative Efficiency

3. Total Surplus is Maximized

- Producing Q_1 units maximizes total surplus, combining consumer and producer surplus.
- Any other quantity would reduce the total surplus, decreasing benefits for consumers or producers.
- Q_1 is allocatively efficient as it maximizes the combined surplus of consumers and producers.

© 2025 McGraw Hill Limited

4.2 EXTERNALITIES AND EFFICIENCY LOSSES 1/4

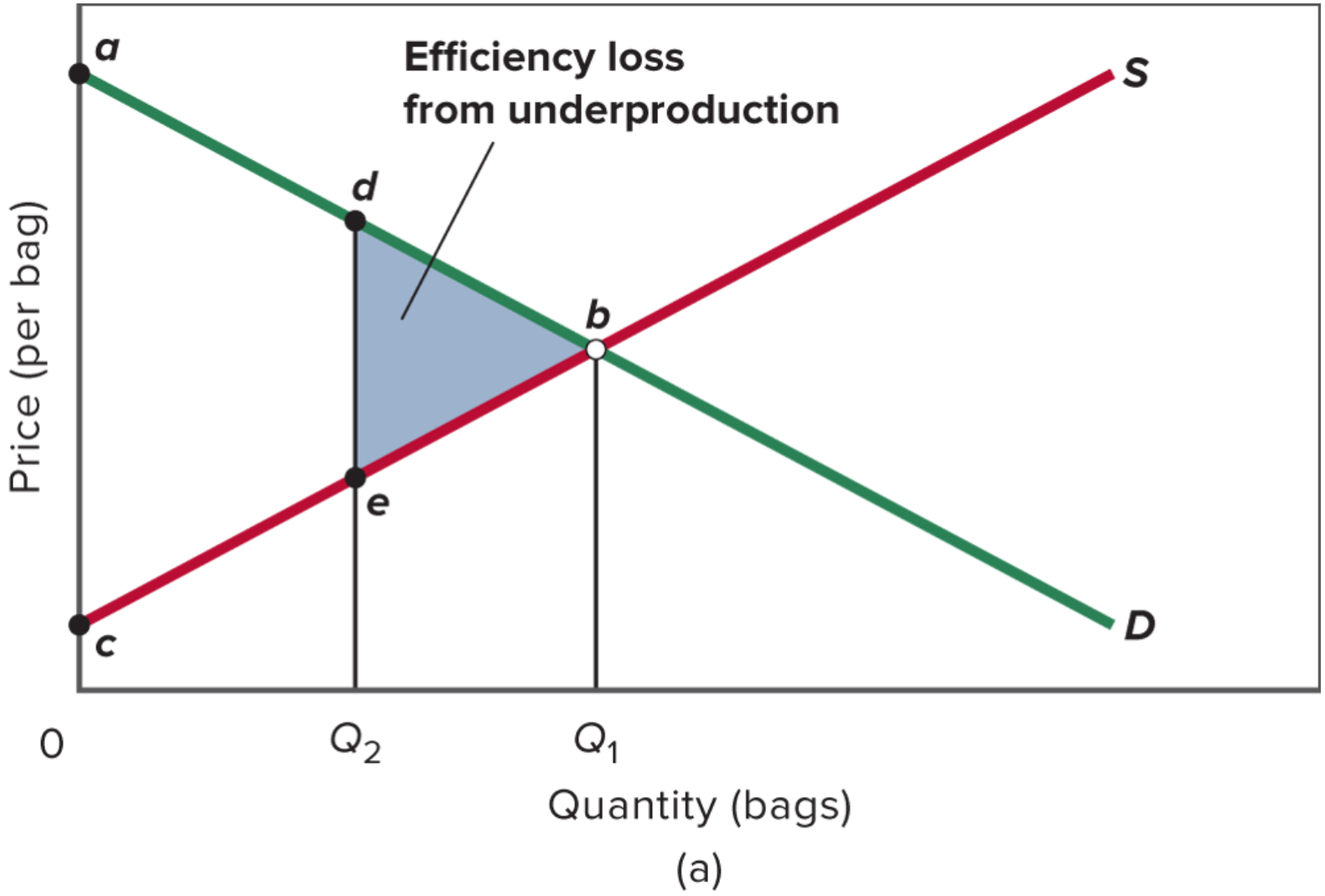
Underproduction, Figure 4-4a.

- Reducing output from Q_1 to Q_2 decreases total surplus, leading to a deadweight loss.
- The grey triangle (dbe) represents the lost surplus when output is below the efficient level.
- From Q_2 to Q_1, consumers are willing to pay more than the cost to producers, creating unrealized net benefits.
- Society loses net benefits when resources are diverted from producing high-value goods to less beneficial alternatives.

© 2025 McGraw Hill Limited

FIGURE 4-4a Efficiency Losses (or Deadweight Losses)

- Quantity levels less than or greater than the efficient quantity Q_1 create efficiency losses.
- Triangle dbe shows the efficiency loss associated with underproduction at output Q_2.



© 2025 McGraw Hill Limited

4.2 EXTERNALITIES AND EFFICIENCY LOSSES 2/4

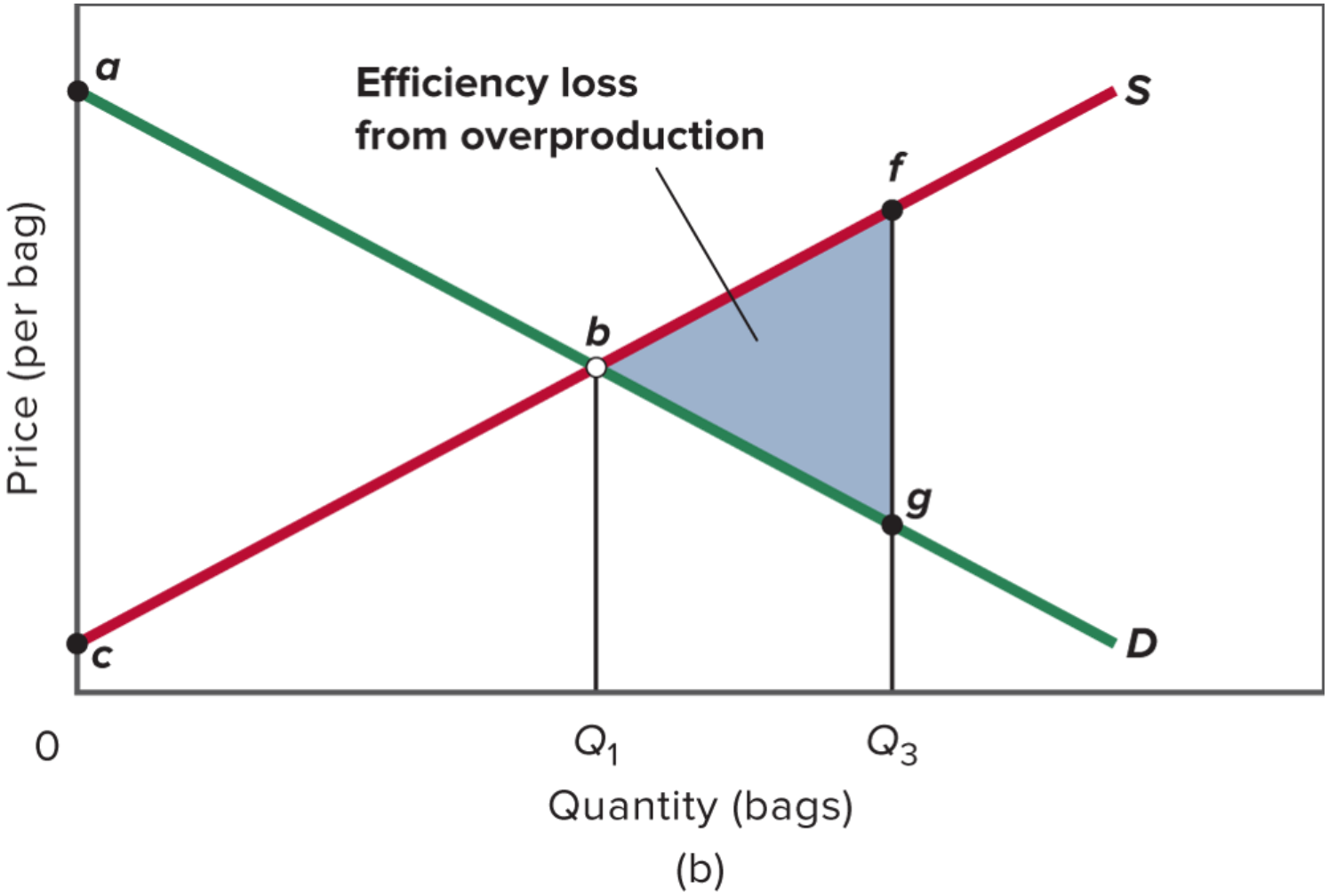
Overproduction, Figure 4-4b.

- Producing beyond Q_1 to Q_3 reduces total surplus by the grey triangle (bfg), causing a deadweight loss.
- From 0 to Q_1, benefits exceed costs, but from Q_1 to Q_3, costs exceed benefits, reducing total surplus.
- Producing beyond Q_1 creates losses because consumers' willingness to pay is less than the production cost.
- Deadweight Loss: Any output level different from Q_1, whether overproduction or underproduction, results in efficiency losses for society.

© 2025 McGraw Hill Limited

FIGURE 4-4b Efficiency Losses (or Deadweight Losses)

- Quantity levels less than or greater than the efficient quantity Q_1 create efficiency losses.
- Triangle **bfg** illustrates the efficiency loss associated with overproduction at output level Q_3.



© 2025 McGraw Hill Limited

4.2 EXTERNALITIES AND EFFICIENCY LOSSES 3/4

- An externality is a cost or benefit accruing to a third party external to the market transaction.

Negative externalities, Figure 4-5a.

- Too much is produced
- Supply-side market failures

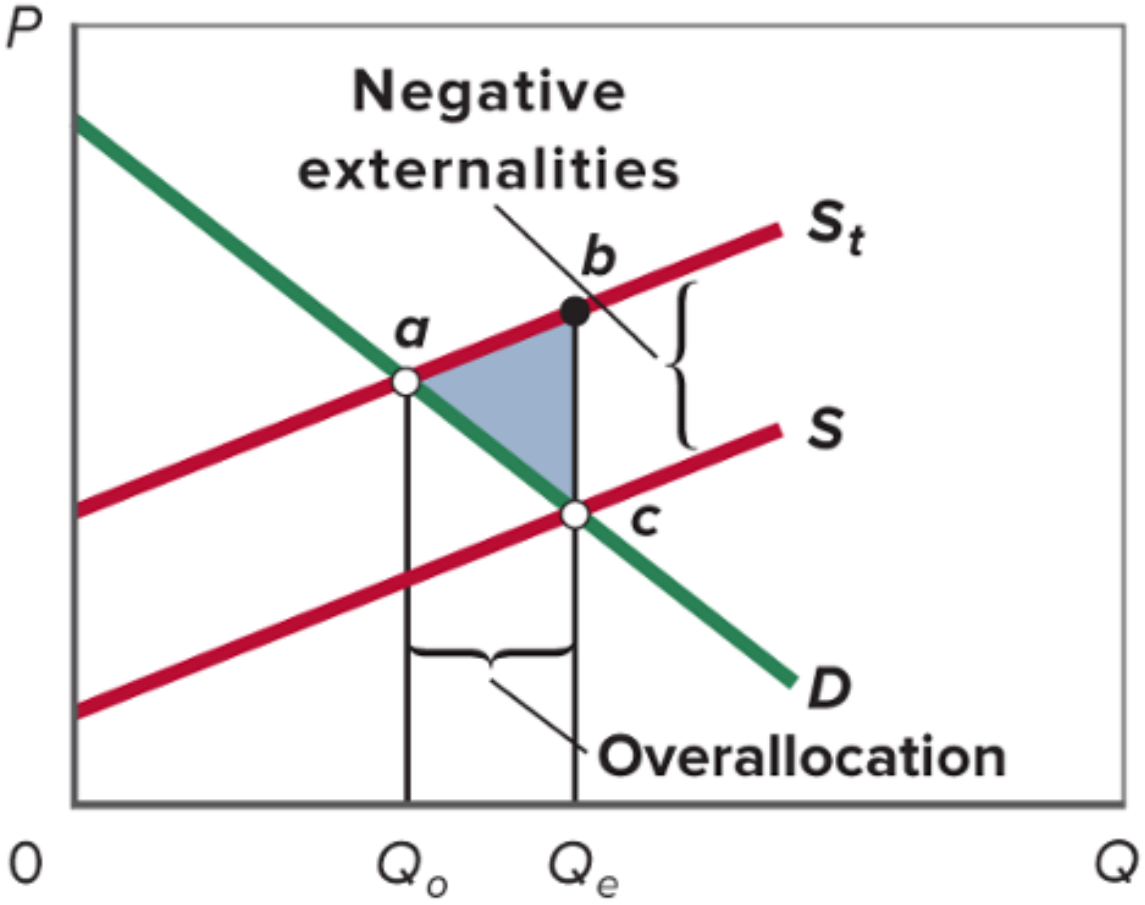
Positive externalities, Figure 4-5b.

- Too little is produced
- Demand-side market failures

© 2025 McGraw Hill Limited

FIGURE 4-5a Negative Externalities and Positive Externalities

- With negative externalities borne by society, the producers' supply curve S is to the right of (below) the full-cost curve S_t.
- Consequently, the equilibrium output Q_e is greater than the optimal output Q_o, and the efficiency loss is abc.

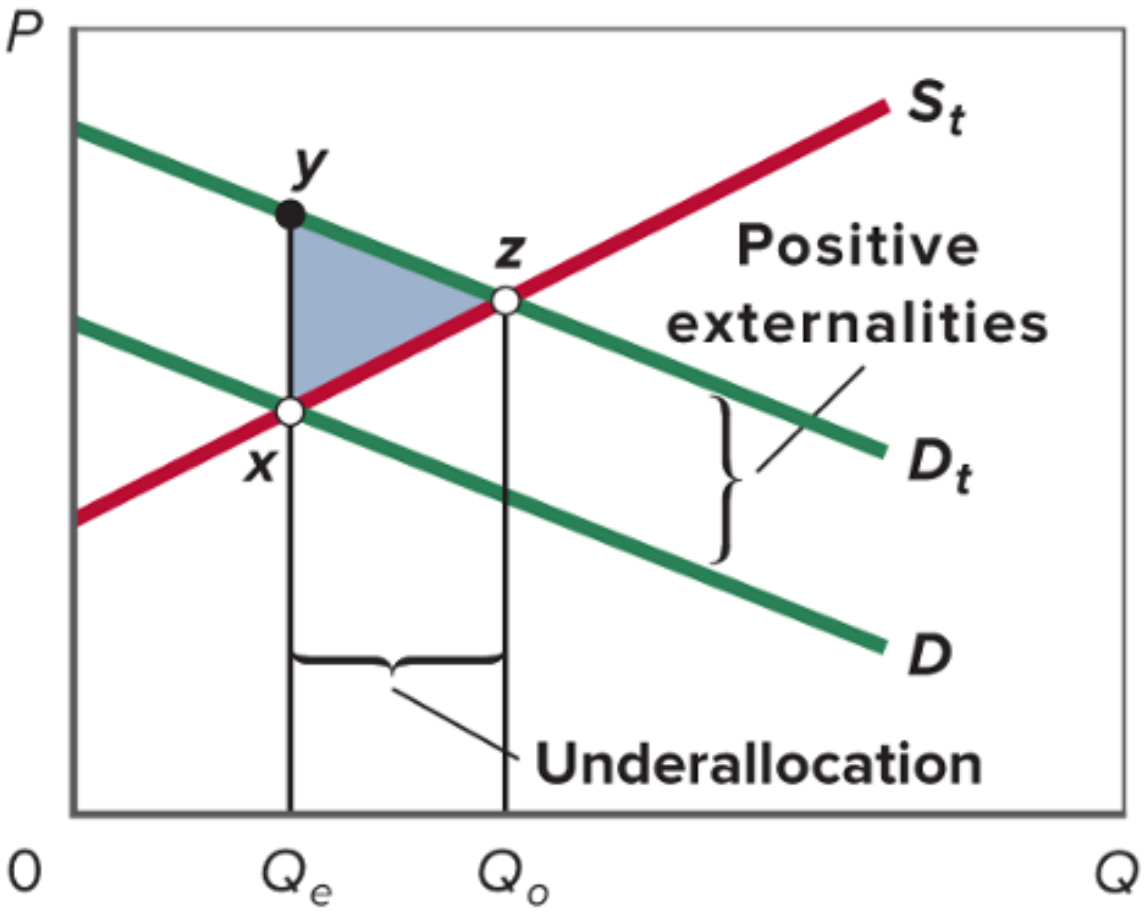


(a)

Negative externalities

FIGURE 4-5b Negative Externalities and Positive Externalities

- When positive externalities accrue to society, the market demand curve D is to the left of (below) the full-benefit demand curve D^t.
- As a result, the equilibrium output Q^e is less than the optimal output Q_o, and the efficiency loss is xyz.



(b)

Positive externalities

4.2 EXTERNALITIES AND EFFICIENCY LOSSES 4/4

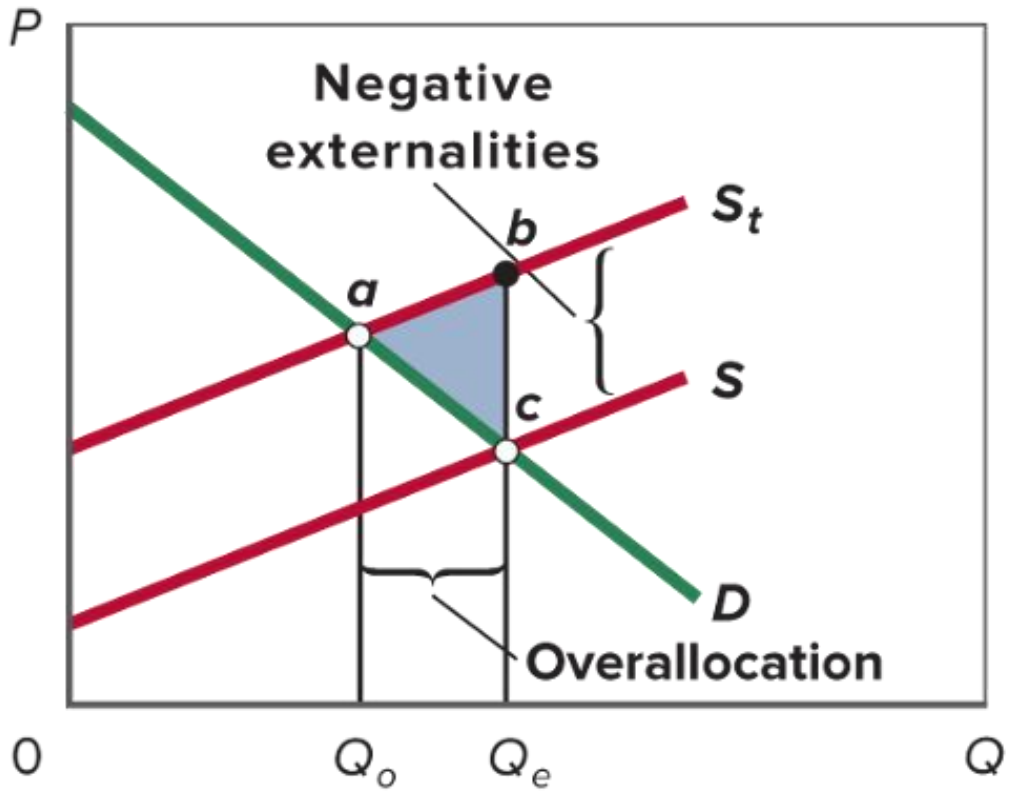
Government intervention

- Correct negative externalities
  - Direct controls: Pass legislation limiting that product.
  - Pigovian tax: Levy taxes or charges specifically on the product.
- Correct positive externalities
  - Subsidies: Subsidies to Buyers and/or Producers of the product.
  - Government provision: It provides the product for free to everyone.

© 2025 McGraw Hill Limited

FIGURE 4-6a Correcting for Negative Externalities

- Negative externalities result in an overallocation of resources.

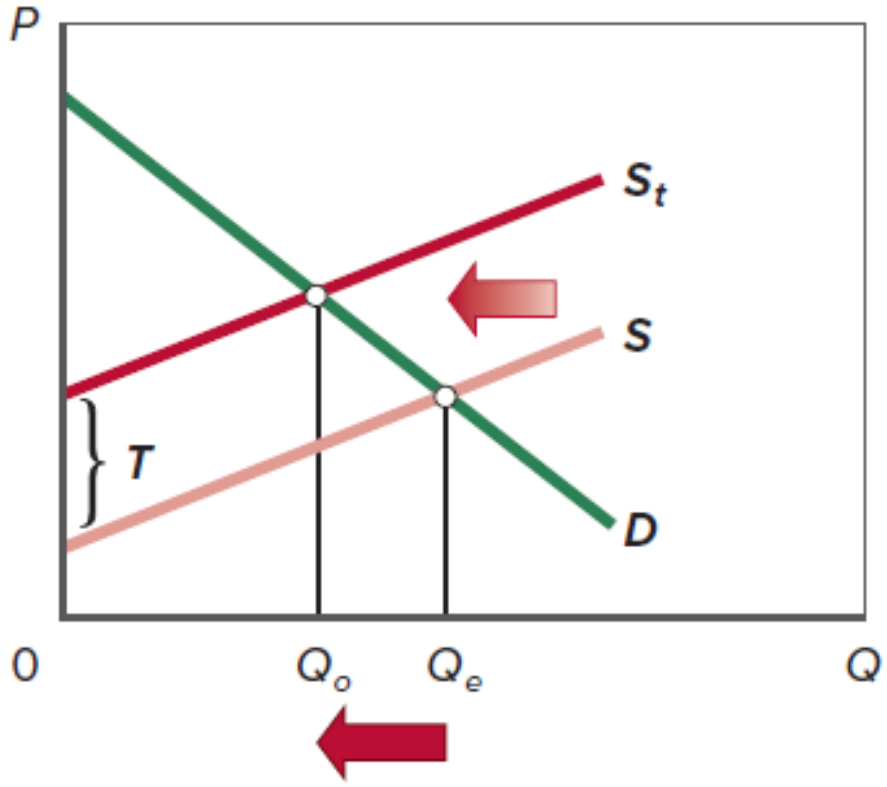


(a)
Negative externalities

FIGURE 4-6b Correcting for Negative Externalities

(1) Use **direct controls**, which would shift the supply curve from S to St and reduce output from Q_e to Q_o,

(2) Impose a **specific tax**, T, that would also shift the supply curve from S to S_t, eliminating the overallocation of resources and, thus, the efficiency loss.

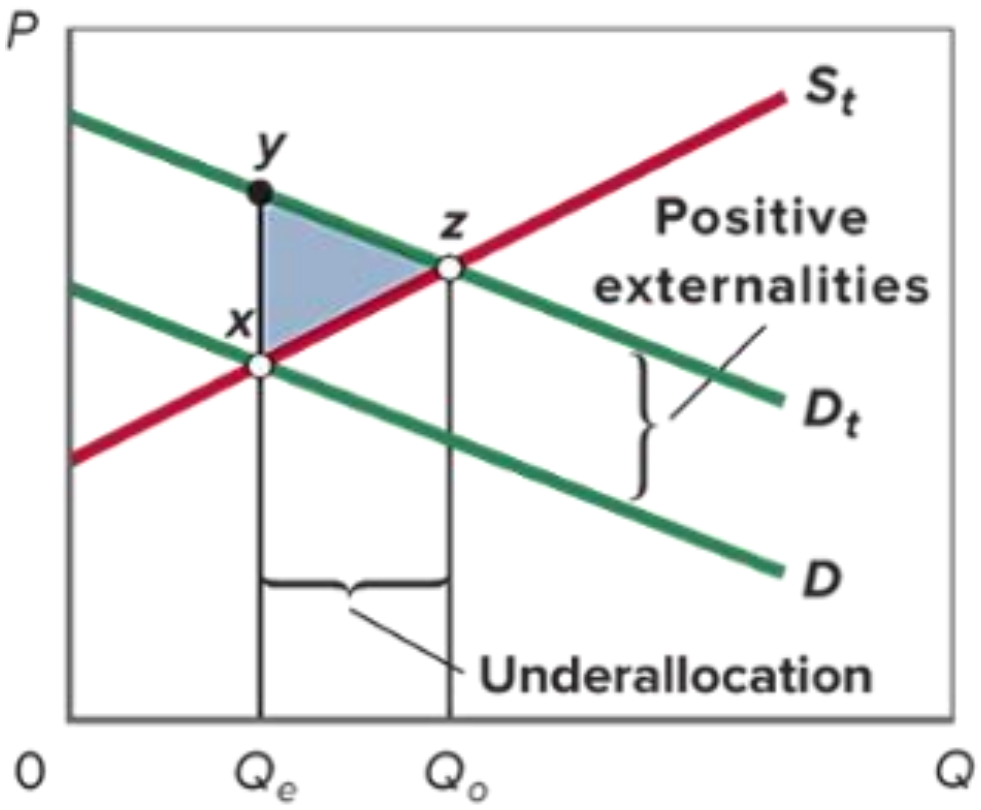


(b)
Correcting the overallocation of resources via direct controls or via a tax

© 2025 McGraw Hill Limited

FIGURE 4-7a Correcting for Positive Externalities

- Positive externalities result in an underallocation of resources.

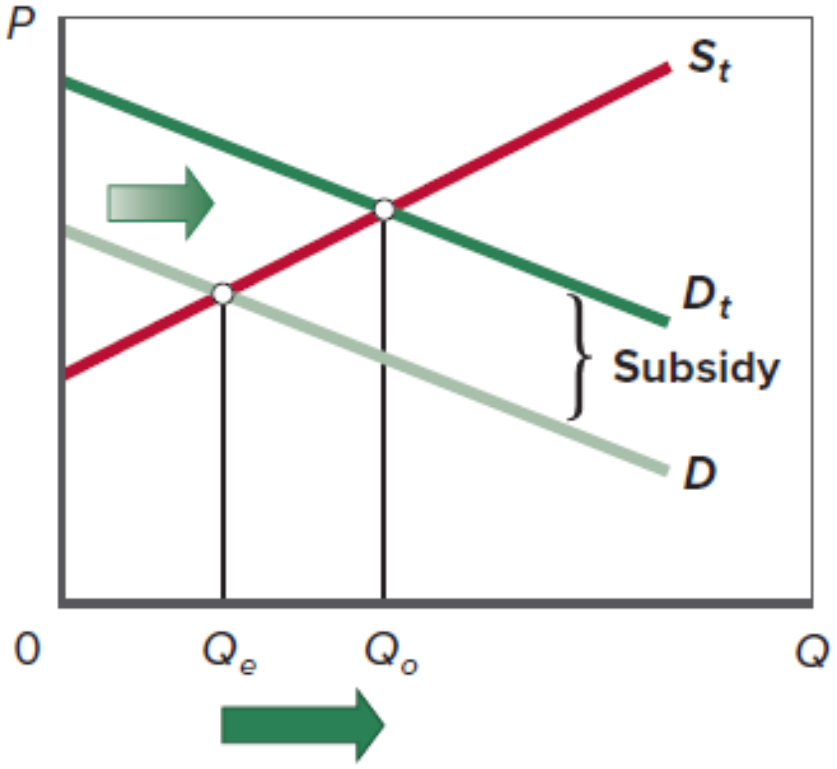


(a)
Positive externalities

© 2025 McGraw Hill Limited

FIGURE 4-7b Correcting for Positive Externalities

- This can be corrected by a **subsidy to consumers**, which shifts market demand from D to D_t and increases output from Q_e to Q_o.

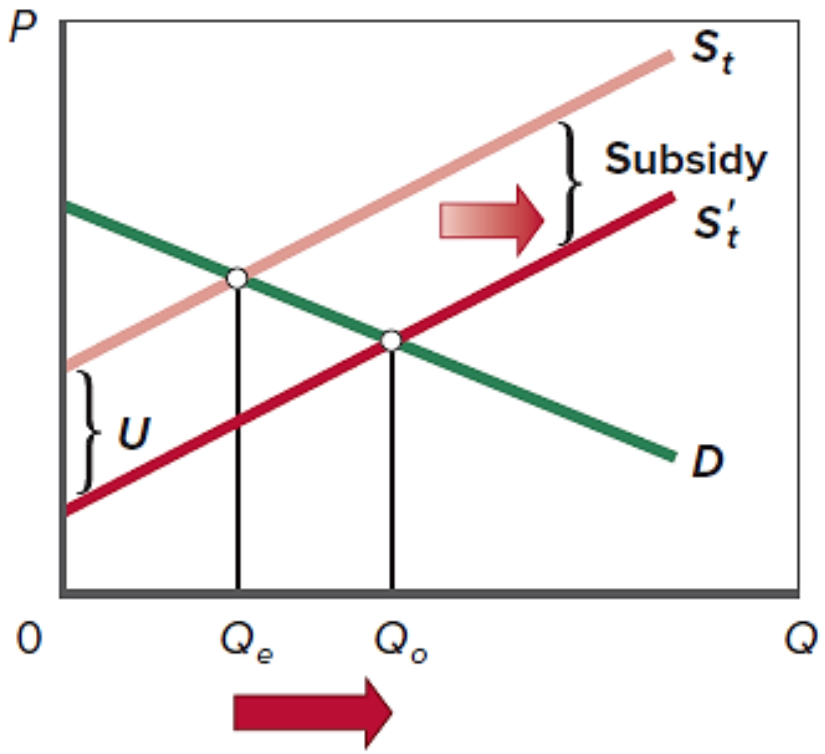


(b)
Correcting the underallocation of resources via a subsidy to consumers

© 2025 McGraw Hill Limited

FIGURE 4-7c Correcting for Positive Externalities

- This can be done by providing **producers a subsidy** of U.
- This shifts their supply curve from S_t to S'_t, increasing output from Q_e to Q_o.



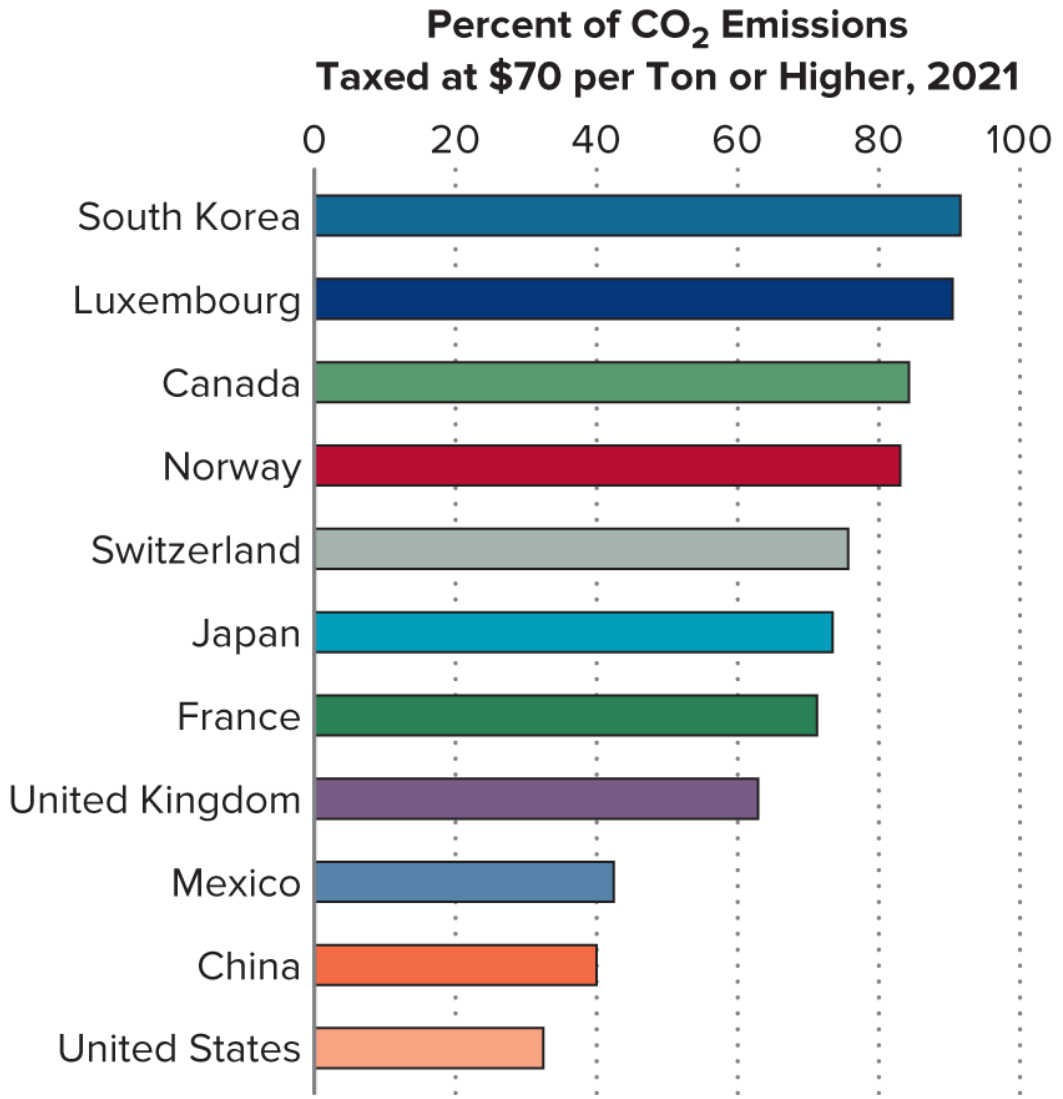
(c)
Correcting the underallocation of resources via a subsidy to producers

© 2025 McGraw Hill Limited

GLOBAL PERSPECTIVE 4.1

Percentage of Carbon Dioxide (CO_2) Emissions Taxed, Selected Nations, 2021

- Countries differ significantly in the percentage of their CO_2 emissions taxed at US$70 per ton (1 ton = 1.01605 tonnes) or higher.
- This variation stems from differences in tax rates and which industries (e.g., agriculture, industry, transportation) are taxed in each country.



Source: Effective Carbon Rates 2021, Organization for Economic Cooperation and Development (OECD), oecd.org.

© 2025 McGraw Hill Limited

TABLE 4-3: Methods for Dealing With Externalities 1/2

Problem	Resource allocation outcome	Ways to correct
Negative externalities (spillover costs)	Overproduction of output and therefore overallocation of resources	1. Private bargaining 2. Liability rules and lawsuits 3. Tax on producers 4. Direct controls 5. Market for externality rights

© 2025 McGraw Hill Limited

TABLE 4-3: Methods for Dealing With Externalities 2/2

Problem	Resource allocation outcome	Ways to correct
Positive externalities (spillover benefits)	Underproduction of output and therefore underallocation of resources	1. Private bargaining 2. Subsidy to consumers 3. Subsidy to producers 4. Government provision

© 2025 McGraw Hill Limited

4.3 SOCIETY'S OPTIMAL AMOUNT OF EXTERNALITY REDUCTION 1/6

Choosing a little more or less

- Negative externalities like pollution are costly to eliminate entirely, despite their harmful effects.
- The goal should be to reduce pollution to an optimal level where marginal costs equal marginal benefits.
- Costs of pollution reduction rise with each additional cleanup effort due to diminishing returns.
- Over-reduction of pollution can harm society by increasing costs more than the benefits.

© 2025 McGraw Hill Limited

4.3 SOCIETY'S OPTIMAL AMOUNT OF EXTERNALITY REDUCTION 2/6

MC, MB, and Equilibrium Quantity

- Downsloping marginal-benefit curve, MB, for pollution reduction (diminishing marginal utility)
- Rising marginal-cost curve, MC, for pollution reduction (increasing opportunity costs)

© 2025 McGraw Hill Limited

4.3 SOCIETY'S OPTIMAL AMOUNT OF EXTERNALITY REDUCTION 3/6

MC, MB, and Equilibrium Quantity

- The MB curve for pollution reduction slopes downward due to diminishing marginal utility.
  - Each additional unit of reduction provides less benefit.
- The MC curve slopes upward because of increasing opportunity costs.
  - Reducing pollution more becomes increasingly expensive.

© 2025 McGraw Hill Limited

4.3 SOCIETY'S OPTIMAL AMOUNT OF EXTERNALITY REDUCTION 4/6

MC, MB, and Equilibrium Quantity

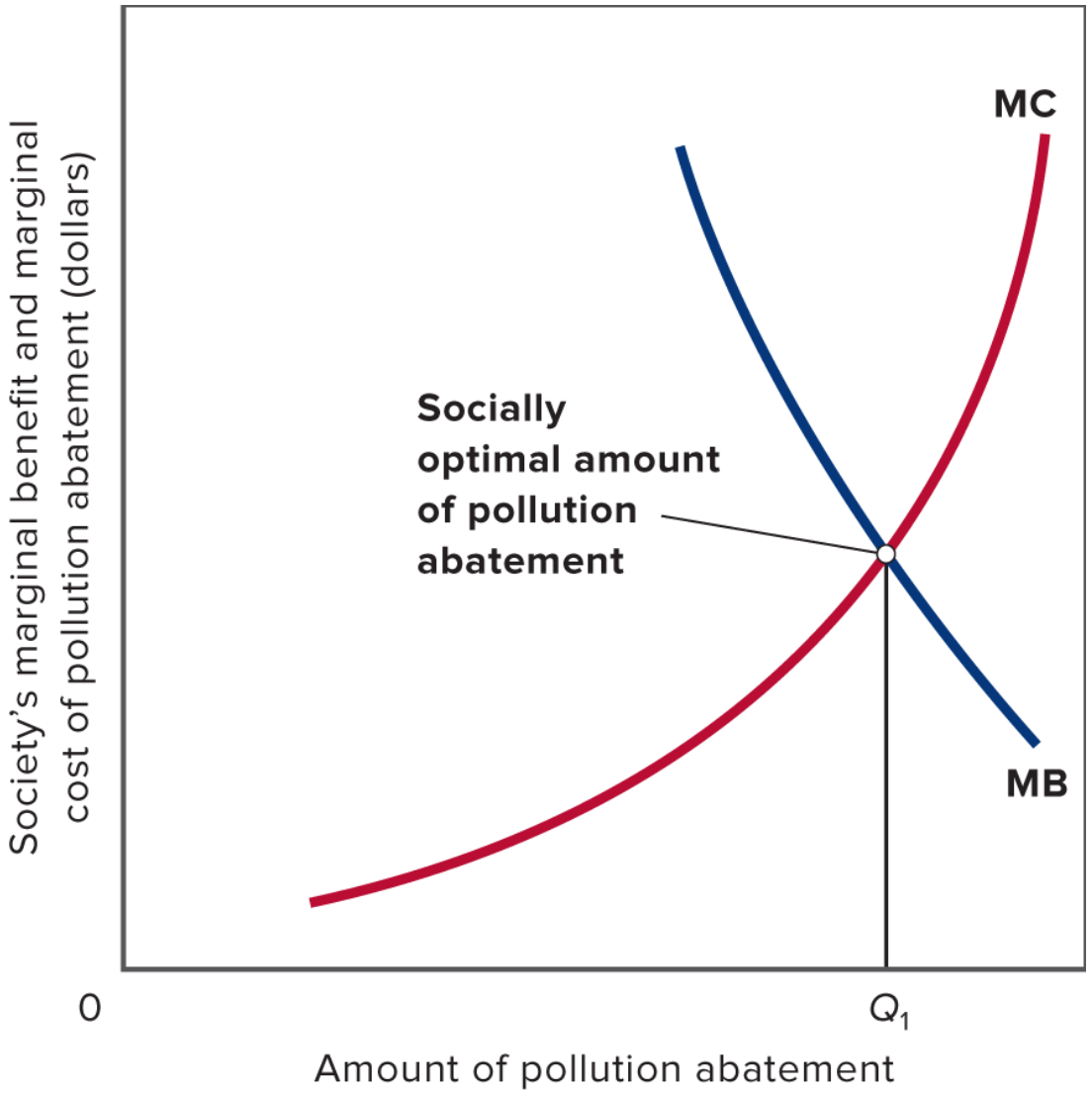
- The optimal reduction of an externality occurs when society's MC = MB.
- Measuring the marginal costs and benefits of pollution control is challenging, but achieving zero pollution may not be economically efficient.
- Allowing some pollution can be better for society if the resources saved provide greater benefits when used elsewhere.

© 2025 McGraw Hill Limited

FIGURE 4-8 Society's Optimal Amount of Pollution Abatement

- The optimal level of pollution abatement is at Q_1, where society's MC equals the marginal MB of reducing the externality.

- At Q_1, the balance between costs and benefits maximizes societal welfare, ensuring resources are used most efficiently for pollution control.



© 2025 McGraw Hill Limited

4.3 SOCIETY'S OPTIMAL AMOUNT OF EXTERNALITY REDUCTION 5/6

Shifts of MB and MC Curves

- Marginal-cost and marginal-benefit curves can shift over time due to technological advancements or changes in societal preferences.
- Improved pollution-control technology lowers costs, shifting the MC curve rightward and increasing optimal abatement.
- New health information may shift the MB curve rightward, raising the optimal level of pollution control.

© 2025 McGraw Hill Limited

4.3 SOCIETY'S OPTIMAL AMOUNT OF EXTERNALITY REDUCTION 6/6

Government's Role When Externalities Are Present

- Correcting market failures is challenging, requiring accurate identification and diagnosis.
- Political pressures often lead to overregulation or underregulation.
- Government actions may prioritize political interests over economic efficiency.
- Government failure occurs when public sector shortcomings result in inefficiency.

© 2025 McGraw Hill Limited

4.4 ASYMMETRIC INFORMATION 1/11

- Asymmetric information is a subtler type of market failure in which one party has private information unavailable to the other.
- This information imbalance can lead to inefficient allocation of resources.
- Government intervention may be needed to equalize information or provide goods/services impacted by this imbalance.
- Addressing asymmetric information can help improve market efficiency and trust.

© 2025 McGraw Hill Limited

When Sellers Possess Private Information

- Sellers with private information can cause market failure by creating an imbalance in transactions.
- This imbalance leads to an underallocation of resources to certain goods or services.
- The lack of accessible information for buyers results in inefficiencies in the market.
- Intervention may be necessary to correct these inefficiencies and ensure better resource allocation.

© 2025 McGraw Hill Limited

4.4 ASYMMETRIC INFORMATION 3/11

When Sellers Possess Private Information: Gasoline Market

- Suppose no system of weights and measures is established by law, no government inspection of gasoline pumps, and no law against false advertising.
- Because of the high cost of verifying information about the seller, many consumers would opt out of this chaotic market.
- The government has established a system of weights and measures and has inspectors.

© 2025 McGraw Hill Limited

4.4 ASYMMETRIC INFORMATION 4/11

When Sellers Possess Private Information: Licensing of surgeons

- Suppose anyone can hang out a shingle and claim to be a surgeon.
- The market's slow correction would cause underallocation to surgery due to high risks.
- Government licensing ensures reliable, low-cost information for critical services.
- Regulations in other sectors increase certainty and provide essential risk warnings.

© 2025 McGraw Hill Limited

4.4 ASYMMETRIC INFORMATION 5/11

When Buyers Possess Private Information

- Buyers with private information can exploit sellers, leading to market failure.
- This information imbalance causes sellers to hesitate, reducing market activity.
- The result is an underallocation of resources to the affected goods or services.
- Reduced market interaction occurs due to sellers' fear of being exploited.

© 2025 McGraw Hill Limited

4.4 ASYMMETRIC INFORMATION 6/11

When Buyers Possess Private Information: Moral Hazzard

- Moral hazard arises when people take more risks after securing a contract that shifts costs to another party.
  - Examples include reckless driving after obtaining insurance or risky loans after securing government guarantees.
- Individuals take risks knowing others will bear the consequences.
- Markets may underallocate resources due to unpredictable buyer behaviour.

© 2025 McGraw Hill Limited

4.4 ASYMMETRIC INFORMATION 7/11

When Buyers Possess Private Information: Moral Hazzard

- Sellers struggle to identify which buyers will likely engage in costly behaviour, leading to market inefficiency.
  - Divorce insurance is an example where moral hazard could lead to more divorces, making the insurance unprofitable.
  - High premiums to offset the risk deter buyers, resulting in a failed market for such insurance.
- Society addresses some moral hazard issues through laws like child support and alimony, which mitigate economic hardship.

© 2025 McGraw Hill Limited

4.4 ASYMMETRIC INFORMATION 8/11

When Buyers Possess Private Information: Adverse Selection

- Adverse selection occurs when high-risk individuals are more likely to buy insurance.
- This skews the insurance pool, leading to higher premiums and driving low-risk individuals away.
- The imbalance increases costs, making insurance less profitable and less attractive.

© 2025 McGraw Hill Limited

4.4 ASYMMETRIC INFORMATION 9/11

When Buyers Possess Private Information: Adverse Selection

- Charging different rates for high-risk and low-risk individuals is impractical due to private information.
- Governments can mitigate adverse selection by mandating universal insurance participation.
- Universal participation balances the insurance pool, including high-risk and low-risk individuals.
- In Canada, mandatory driver insurance and government-provided health care prevent adverse selection.

© 2025 McGraw Hill Limited

4.4 ASYMMETRIC INFORMATION 10/11

Qualification

- Not all information asymmetries require government intervention.
- Households and businesses often develop solutions, like product warranties and franchising, to overcome these issues.
- Private firms and organizations provide valuable information through services like Consumer Reports and credit bureaus.

© 2025 McGraw Hill Limited

4.4 ASYMMETRIC INFORMATION 11/11

Qualification

- Information from brokers, bonding agencies, and intermediaries also helps mitigate asymmetry.
- However, the private sector cannot solve all information problems.
- Government intervention is sometimes necessary for efficient resource allocation.

© 2025 McGraw Hill Limited

THE LAST WORD 1/2

Visible Pollution, Hidden Costs

How can governments reduce air pollution at the lowest possible cost if only the polluters themselves know the costs of abatement?

- Outright fossil fuel bans are costly and disruptive; alternatives like carbon taxes are more efficient.
- Effective pollution control requires balancing marginal costs and benefits.

Nordroden/123RF

© 2025 McGraw Hill Limited

THE LAST WORD 2/2

Visible Pollution, Hidden Costs

- Government faces challenges due to asymmetric information on abatement costs.
- Tradeable emissions permits ("cap and trade") create market-driven pollution reduction.
- High-cost polluters buy permits from low-cost polluters, optimizing emissions reduction.
- Emissions trading achieves pollution reduction at the lowest possible cost.

© 2025 McGraw Hill Limited

CHAPTER SUMMARY 1/4

- Consumer Surplus is the difference between what consumers are willing to pay and what they pay.
- Producer surplus is the difference between the minimum price producers accept and the higher price they receive.
- Total Surplus is the sum of consumer and producer surplus.
- Total surplus is maximized at equilibrium, where marginal benefit equals marginal cost.

© 2025 McGraw Hill Limited

CHAPTER SUMMARY 2/4

- Efficiency losses occur when output is less than or greater than equilibrium, leading to deadweight losses due to underproduction or overproduction.
- Negative externalities cause an overallocation of resources, as market prices do not fully capture costs, leading to overproduction.
- Positive externalities lead to underallocation of resources, as benefits are not fully reflected in market demand, resulting in underproduction.

© 2025 McGraw Hill Limited

CHAPTER SUMMARY 3/4

- Direct controls and Pigovian taxes are used to correct negative externalities by increasing production costs, reducing the externality and the overallocation of resources.
- The government can subsidize consumers or producers to correct underallocation caused by positive externalities, increasing equilibrium output.
- Coase Theorem suggests that private bargaining can sometimes resolve externalities without government intervention.

© 2025 McGraw Hill Limited

CHAPTER SUMMARY 4/4

- Optimal Externality Abatement is achieved when the marginal cost of reducing the externality equals the marginal benefit, typically less than complete elimination.
- Asymmetric Information justifies government intervention when unequal information causes market failure, leading to underallocation of resources.
- Moral hazard and adverse selection problems arise from asymmetric information, where one party's behaviour or information advantage leads to costly outcomes for the other party.

© 2025 McGraw Hill Limited